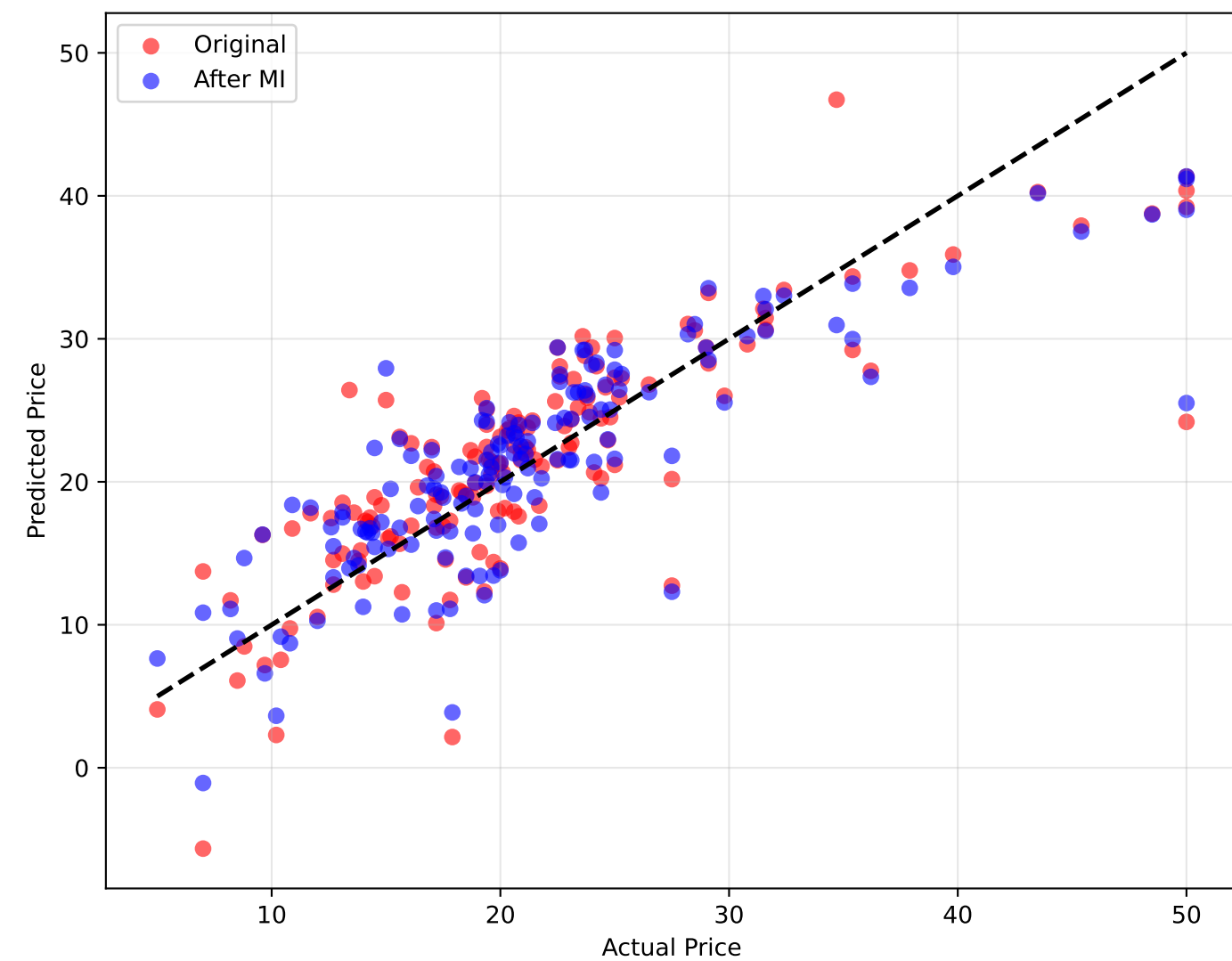
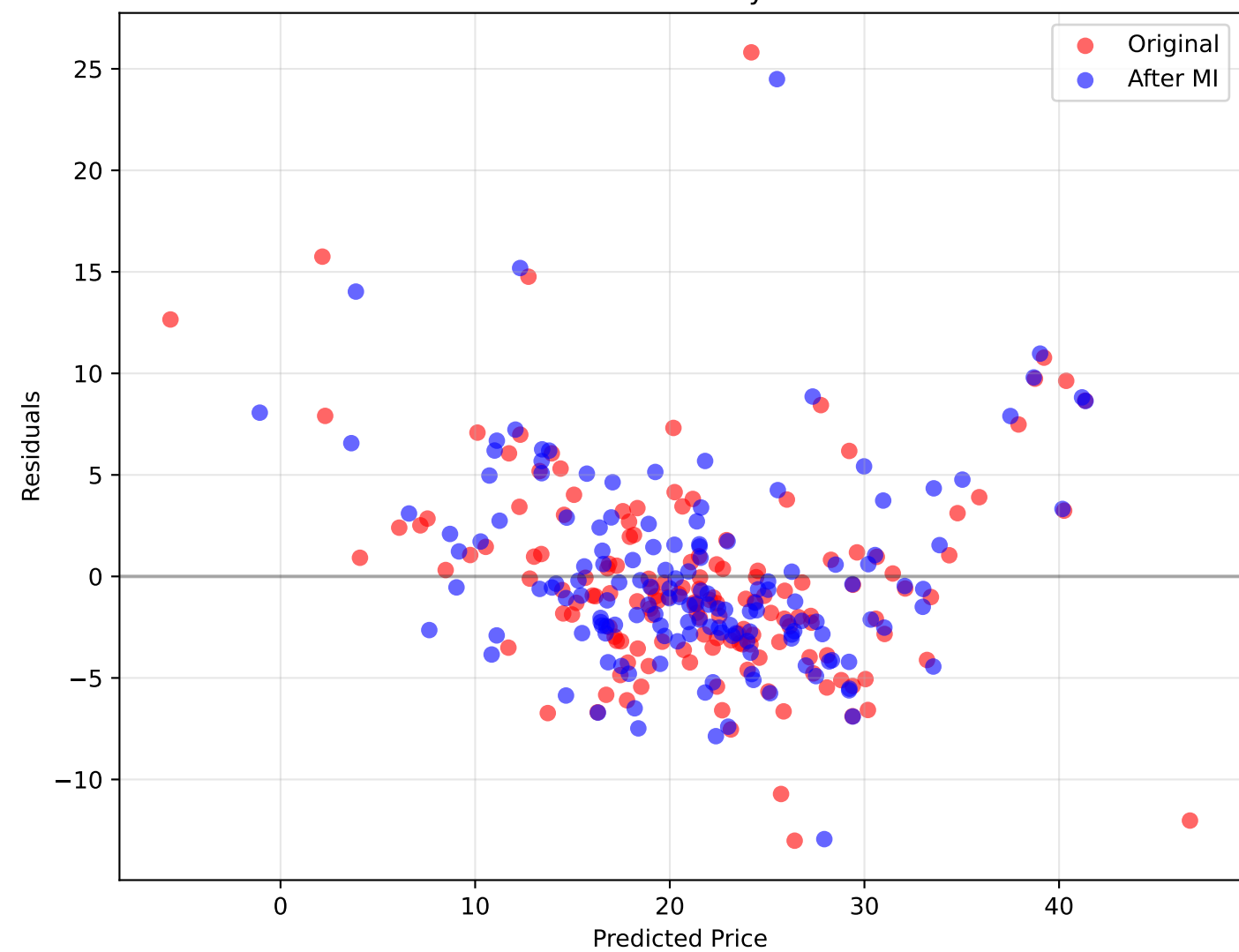


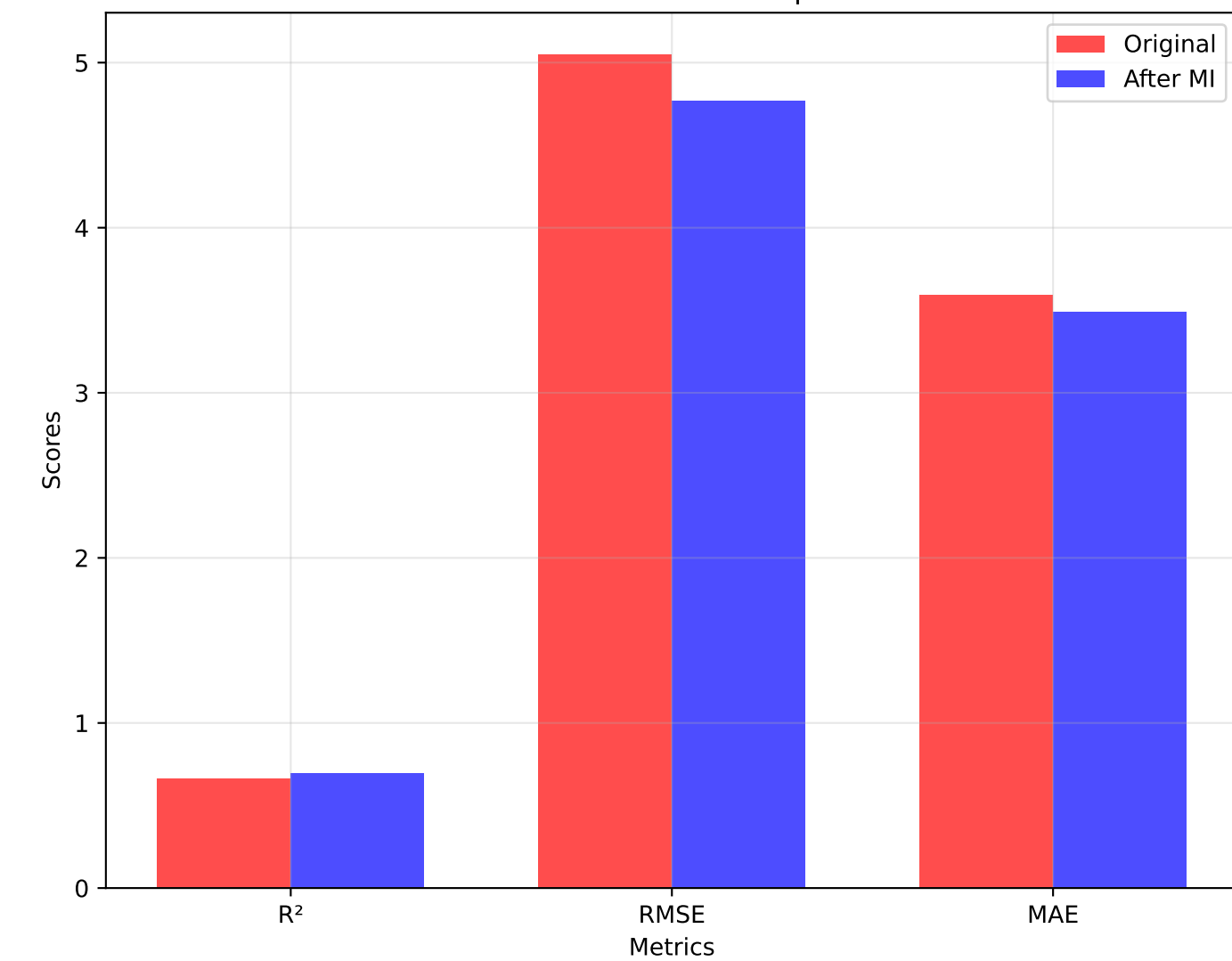
Prediction vs Actual



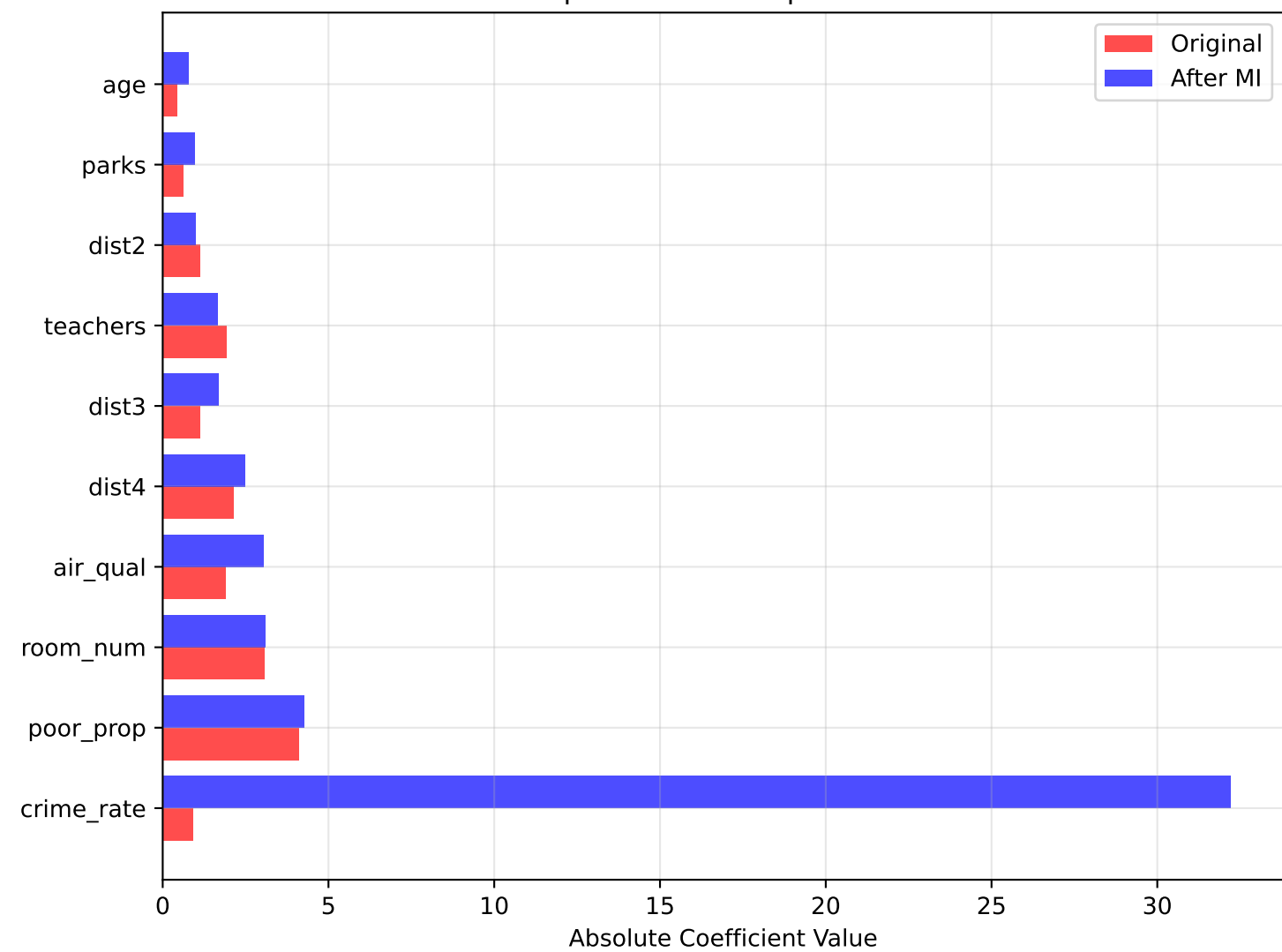
Residual Analysis



Model Performance Comparison



Top 10 Feature Importance



REGRESSION ANALYSIS SUMMARY

REGRESSION MODEL COMPARISON SUMMARY

ORIGINAL DATA MODEL:

- R^2 Score: 0.6594
- RMSE: 5.0498
- MAE: 3.5936
- Residual Std: 5.0442

AFTER MULTIPLE IMPUTATION MODEL:

- R^2 Score: 0.6966
- RMSE: 4.7657
- MAE: 3.4870
- Residual Std: 4.7656

IMPROVEMENT:

- ΔR^2 : +0.0372
- Δ RMSE: +0.2841
- Δ MAE: +0.1067

TOP 5 PRICE DRIVERS (After MI):

1. crime_rate: 32.2095
2. poor_prop: 4.2738
3. room_num: 3.1064
4. air_qual: 3.0341
5. dist4: 2.4835

INTERPRETATION:

The multiple imputation approach improves model performance while providing more robust coefficient estimates.